

minimal langgraph mvp

Minimal LangGraph Todo MVP

Goal

Build a small LangGraph workflow that processes transcribed audio and updates a todo list.

Keep STT outside LangGraph:

Audio input -> Speech-to-text -> LangGraph -> Todo tools -> Database -> Text/UI response

This is not a multi-agent system. It is a single LangGraph agent/workflow with tools.

Minimal LangGraph Nodes



Nodes Needed

Node	Purpose
'classify_intent'	Decide what the user wants: add, update, complete, list, or clarify.
'add_todo'	Extract tasks from transcript and save them.
'update_todo'	Change priority, due date, title, or notes.
'complete_todo'	Match spoken completed task to existing todo and mark it completed.
'list_todos'	Fetch today's pending/completed todos.
'clarify'	Ask a short question when task or detail is unclear.
'generate_response'	Return simple text for UI.

Minimal Tools

Tool	Purpose
'create_todo(title, due_date, priority)'	Add new todo.
'list_todos(date, status)'	Get todos.
'update_todo(todo_id, fields)'	Change priority/details/status.
'complete_todo(todo_id, note)'	Mark completed and create time log.
'find_matching_todo(text)'	Fuzzy match spoken task to saved todo.

Minimal Intents

ADD_TODO
UPDATE_TODO
COMPLETE_TODO
LIST_TODOS
CLARIFY

Simple Flows

Add Todo

"I need to fix loan API today"
-> classify_intent: ADD_TODO
-> add_todo
-> create_todo()
-> response: "Added: Fix loan API."

Update Priority

"Make loan API high priority"
-> classify_intent: UPDATE_TODO
-> update_todo
-> find_matching_todo()
-> update_todo()
-> response: "Updated priority to high."

Complete Todo

```
"I completed the loan API task"  
-> classify_intent: COMPLETE_TODO  
-> complete_todo  
-> find_matching_todo()  
-> complete_todo()  
-> response: "Marked loan API as completed."
```

List Todos

```
"What is left today?"  
-> classify_intent: LIST_TODOS  
-> list_todos()  
-> response: pending todos
```

Keep It Simple

- Use one LangGraph workflow.
- Use tools for database actions.
- Use fuzzy matching only when updating or completing existing todos.
- Ask clarification only when task matching or required details are unclear.
- Do not add TTS/audio response for the MVP.